

# Milestone 2 Complete Guide

## Objective

Build a complete Certificate & Permit Management System where:

- Citizens submit applications
  - Officials verify documents
  - Departments approve/reject requests
  - Certificates are generated automatically
  - Citizens download certificates
  - Every step is tracked
- 

## Users (Roles)

### 1. Citizen

- Login/Register
  - Apply for Certificate
  - Upload Documents
  - Track Application
  - Download Certificate
- 

### 2. Verification Officer

- View Pending Applications
  - Verify Documents
  - Request Re-upload
  - Forward to Approver
- 

### 3. Department Officer

- Approve Application
  - Reject Application
  - Add Remarks
- 

#### **4. Admin**

- Manage Certificate Types
  - Manage Permit Types
  - Manage Departments
  - View Reports
- 

## **Certificates**

Implement these:

- Birth Certificate
  - Death Certificate
  - Income Certificate
  - Residence Certificate
  - Marriage Certificate
- 

## **Permits**

Implement:

- Trade License
  - Shop License
  - Building Permit
  - Water Connection Permit
- 

## **Database Design**

## Citizen

citizen\_id  
name  
email  
phone  
aadhaar  
address  
ward

---

## Application

application\_id  
application\_no  
citizen\_id  
certificate\_type  
status  
submission\_date  
approval\_date  
department

Status

Submitted

Under Verification

Verified

Approved

Rejected

Certificate Generated

---

## Documents

document\_id  
application\_id  
document\_name

document\_url  
verification\_status  
remarks

---

## Certificate

certificate\_id  
certificate\_no  
application\_id  
issue\_date  
digital\_signature  
pdf\_url

---

## Workflow

Citizen Login



Select Certificate



Fill Form



Upload Documents



Submit Application



Verification Officer



Verify Documents

↓

Approved

↓

Generate Certificate

↓

Citizen Download

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## Backend Microservice

Create one microservice

service-management-service

Modules

controller

service

repository

entity

dto

mapper

exception

config

security

---

# REST APIs

## Citizen APIs

### Apply Certificate

POST

/api/certificates/apply

---

### Get My Applications

GET

/api/certificates/my

---

### Get Application Details

GET

/api/certificates/{id}

---

### Download Certificate

GET

/api/certificate/download/{id}

---

## Officer APIs

### Pending Applications

GET

/api/officer/pending

---

Verify Documents

PUT

/api/officer/verify/{id}

---

Approve

PUT

/api/officer/approve/{id}

---

Reject

PUT

/api/officer/reject/{id}

---

Generate Certificate

POST

/api/certificate/generate/{id}

---

# React Pages

## Citizen

### Login

/login

---

## Dashboard

Cards

Total Applications

Approved

Pending

Rejected

---

## Apply Certificate

Form

Certificate Type

Applicant Name

DOB

Address

Phone

Email

Upload Documents

Submit

---

## My Applications

Table

Application No

Certificate

Status



Date

Action

Action

View

Download

---

## Track Application

Timeline

Submitted

↓

Verification

↓

Approval

↓

Certificate Generated

---

## Officer Pages

Dashboard

Cards

Pending

Verified

Rejected

Generated

---

Pending Applications

Table

Application No

Citizen

Certificate

Status

Action

Buttons

Verify

Approve

Reject

---

Verification Page

Left Side

Citizen Details

Right Side

Uploaded Documents

Buttons

Verified

Need Correction

---

Approval Page

Application Details

Documents

Remarks

Buttons

Approve

Reject

---

Certificate Preview

Show PDF

Buttons

Generate

Digital Sign

Download

---

## Admin Pages

Manage Certificates

Add

Edit

Delete

---

Departments

Revenue

Municipality

Water

Housing

Education

---

Reports

Charts

Applications

Approval %

Rejected %

Monthly Certificates

---

## Validation

Citizen

Mandatory Fields

Email Validation

Phone Validation

Aadhaar Validation

DOB Validation

---

Documents

Only PDF

Only JPG

Max 5MB

---

Approval

Cannot approve

unless

Verification Complete

---

Certificate

Cannot Generate

until

Approval Complete

---

## Dashboard Metrics

Citizen Dashboard

Applications

Approved

Rejected

Pending

Officer Dashboard

Pending Verification

Approved Today

Rejected Today

Certificates Generated

Admin Dashboard

Applications Today

Monthly Certificates

Department Performance

Approval Time

Pending Applications

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## Spring Boot Technologies

- Java 25
- Spring Boot 4
- Spring Security
- Spring Data JPA
- PostgreSQL
- Redis (optional cache)
- Kafka (publish events such as `ApplicationSubmitted`, `DocumentVerified`, `CertificateGenerated`)
- Keycloak for authentication/RBAC
- MapStruct
- Lombok
- OpenAPI (Swagger)

These align with the project's prescribed technology stack and architecture.

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# Screens Required (Milestone 2)

Your project document requires these validation/output screens for Milestone 2: **Application Submission, Document Verification, Approval Workflow, Certificate Generation, Digital Signing, and Download Tracking.**

A complete UI set should include:

1. Login
  2. Citizen Dashboard
  3. Apply Certificate
  4. Upload Documents
  5. My Applications
  6. Track Application
  7. Officer Dashboard
  8. Pending Applications
  9. Document Verification
  10. Approval Workflow
  11. Certificate Preview
  12. Digital Signature
  13. Download Certificate
  14. Admin Dashboard
  15. Certificate Management
  16. Department Management
  17. Reports & Analytics
- 

## Folder Structure

civicpulse/

frontend/

```
| — src/  
|   | — pages/  
|   | — components/  
|   | — layouts/  
|   | — services/  
|   | — hooks/  
|   | — routes/
```

backend/

```
| — service-management-service/
```

```
| — controller/  
| — service/  
| — repository/  
| — entity/  
| — dto/  
| — mapper/  
| — security/  
| — exception/  
| — config/
```

```
database/  
| — schema.sql
```

```
docker/  
| — docker-compose.yml
```

```
kubernetes/  
| — deployment.yaml
```

## Expected Demonstration Flow

1. Citizen logs in.
2. Applies for a Birth Certificate.
3. Uploads Aadhaar and supporting documents.
4. Application status becomes **Submitted**.
5. Verification Officer reviews and verifies the documents.
6. Department Officer approves the application.
7. The system generates a certificate with a unique certificate number.
8. The citizen downloads the generated certificate.
9. The admin dashboard reflects the new application and approval statistics.